

2 Kernels and RKHS

1 Find the RKHS over $\mathbb{R}^d$ for each of the following kernels.

$$K(x, y) = x^\top y + c, \quad x, y \in \mathbb{R}^d, c \in \mathbb{R}^+.$$

Notice that $K(x, y)$ can be written as

$$K(x, y) = (y^\top, \sqrt{c})(x^\top, \sqrt{c})^\top, \quad x', y' \in \mathbb{R}^{d+1}, c \in \mathbb{R}^+$$

Guess that

$$H = \{f_{v, v_0}; f_{v, v_0}(x) = (v')^\top x' = (v^\top, \sqrt{c})(x^\top, \sqrt{c})^\top - c + v_0 \quad v', x' \in \mathbb{R}^{d+1}, v_0 \in \mathbb{R}\}$$

H is a vector space. Let all vectors in $\mathcal{X}$ now be augmented $\in \mathbb{R}^{d+1}$

$$\begin{aligned} f_{(v, v_0) + (w, w_0)} &= f_{(v, v_0)} + f_{(w, w_0)} \text{ since} \\ f_{(v, v_0) + (w, w_0)}(x) &= (v + w)^\top x + v_0 + w_0 = (v^\top + w^\top)x + v_0 + w_0 = f_{(v, v_0)}(x) + f_{(w, w_0)}(x) \\ f_{\alpha(v, v_0)} &= \alpha f_{(v, v_0)} \text{ since } f_{\alpha(v, v_0)}(x) = \alpha v^\top + \alpha v_0 = \alpha f_{(v, v_0)}(x) \end{aligned}$$

$K(x, y)$ is in H.

$$K(x, y) = x^\top y = y^\top x = f_{(y, y_0=0)} \quad x, y \in \mathbb{R}^{d+1}$$

H is also an inner product space.

$$\begin{aligned} \langle f_{v, v_0}, f_{w, w_0} \rangle &= \sum_{ij} \alpha_i \beta_j x_i^\top y_j = \left(\sum_i^n \alpha_i x_i \right)^\top \left(\sum_j^m \beta_j y_j \right) \\ &= v^\top w \end{aligned}$$

Also H has a norm induced from the inner product

$$\|f_{v, v_0}\| = v^\top v = \|v\|^2$$

Since $x \in \mathbb{R}^d$ for finite d, H is complete. Hence H is a Hilbert space.

By this construction of H, it has the reproducing property because

$$\begin{aligned} f_{v, v_0=0}(y) &= v^\top y \implies f_{v, v_0=0} = v^\top \text{ and} \\ \langle f_{v, v_0}, K(y, \cdot) \rangle &= \langle f_{v, v_0}, f_{(y, y_0=0)} \rangle = v^\top y \end{aligned}$$

Thus H is RKHS.

$$2. K(x_i, x_j) = (x_i^\top x_j)^2$$

Notice if we write $x_i^\top x_j = (x_i^\top I x_j)$ where I is the identity matrix, then this is a bilinear form. Any bilinear form can be expressed as a quadratic form; with $x_i \equiv x$ as $K = x_i^\top P^\top I P x_i = x P^\top P x = x^\top A x$, $A \equiv P^\top P$. Now substituting this expression into K gives

$$K(x_i, x_j) = (x^\top A x)(x^\top A x) = x^\top A (x x^\top) A x = x^\top A^\top B A x \text{ with } B \equiv x x^\top$$

Now for $x \in \mathbb{R}^d$, guess that H is

$$\{f_S : f_S(x) = x^\top S x \text{ where } S \text{ is a symmetric } d \times d \text{ matrix with} \\ \langle f_{S_A}, f_{S_B} \rangle \equiv \langle S_A, S_B \rangle = \sum_{ij}^d [S_A]_{ij} [S_B]_{ij}$$

H is a vector space since

$$f_{S_A} + f_{S_B} = f_{S_A + S_B} \text{ since } x^\top S_A x + x^\top S_B x = x^\top (S_A + S_B) x \text{ and} \\ f_{\alpha S} = \alpha f_S \text{ since } x^\top \alpha S x = \alpha x^\top S x$$

H is complete since it is over a space of finite dimensions $d \times d$.

$$K(x, \cdot) \in H \text{ since for } S_B \equiv A^\top B A \text{ we have } k(x_i, x_j) = x_i^\top S_B x_j \implies$$

$$K(x_i, \cdot) = f_{S_B}$$

K also has the reproducing property.

$$\langle f_S, K(x, \cdot) \rangle = \langle f_S, f_{S_B} \rangle = \sum_{ij}^d [S]_{ij} [S_B]_{ij} = \\ \sum_{ij} [S]_{ij} [A^\top B A]_{ij}$$

Now $B = x x^\top$ and is a symmetric pos. definite matrix and so $A^\top B A = B$, takes the form below. If we allow postscripts to represent the components of the vector x , i.e $x = (x^1, x^2, \dots, x^d)$, then $A^\top B A$ takes the form

$$A^\top B A = \begin{bmatrix} x^1 x^1 & x^1 x^2 & \dots & x^1 x^d \\ \vdots & & & \vdots \\ x^d x^1 & x^d x^2 & \dots & x^d x^d \end{bmatrix}$$

The matrix composed of products of the components of the same vector x_i . Then $[B]_{mn} = x^m x^n$ and so

$$\langle f_S, K(x_i, \cdot) \rangle = \sum_{mn}^d [B]_{mn} [S]_{mn} = \sum_{mn}^d x^m x^n [S]_{mn} \Big|_{x=x_i} = x_i^\top S x_i = f_S(x_i)$$

Thus H is a RKHS with the kernel K .

